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DATA WAREHOUSES - Summer - 2025/2026

Bitcoin Blockchain
Analytics Platform

Part 1 · Analysis - Dataset Selection & User Needs

2	4+	6+	5
DATA SOURCES	DIMENSIONS	MEASURES	OLAP NEEDS

4-Stage Data Warehouse Implementation

01 ANALYSIS

- → Dataset selection & profiling
- → User type identification
- → Dimension & measure identification
- → OLAP need specification

Presentation

↑ I AM HERE ↑

02 DESIGN

- → Multidimensional model (star schema)
- → Logical Data Map
- → ETL pipeline design
- → SSAS cube architecture

ERD + Docs

03 IMPLEMENTATION

- → Staging DB creation (SQL Server)
- → ETL with SSIS packages
- → Dimensional DW load
- → SSAS multidimensional cube

SSIS + SSAS

04 USAGE & OLAP

- → Interactive dashboards
- → Slice & dice analysis
- → Trend & comparison reports
- → User need fulfilment

Tableau / Power BI

What I Am Building

Source 1
On-Chain
mempool.space API

Source 2
Market Data
CoinGecko API

Staging DB
SQL Server
Raw relational store

Data Mart
Star Schema
Fact + Dimensions

SSAS Cube
Multidim. cube
OLAP aggregations

Reports
Tableau / Power BI

ETL - SSIS Pipeline

Staging Database

Decouples raw extraction from the cleaned dimensional model. Separate SQL Server database for initial load and cleansing.

Single Data Mart

One business process - Bitcoin TX settlement - as a star schema. Single line of business, manageable scope.

SSAS OLAP Cube

Multidimensional cube enables slice & dice, drill-down, and time-series analysis across all dimensions.

Bitcoin Blockchain Key Concepts

Understanding the domain before modelling the data

Block

A batch of verified transactions permanently appended to the chain. Identified by height (sequential) and hash (fingerprint).

Transaction (TX)

A transfer of BTC value: inputs (coin sources / UTXOs spent) → outputs (new UTXOs). Measured in satoshis (1 BTC = 100M sat).

UTXO

Unspent Transaction Output - the fundamental accounting unit. Bitcoin has no 'accounts', only UTXOs that are created and consumed.

Fee Rate (sat/vByte)

Fee per virtual byte. Miners prioritise TXs with higher fee rates - the key economic signal for congestion analysis.

Halving Epoch

Every 210K blocks, the block subsidy halves. Defines major market cycles and miner revenue structure over time.

Network at a Glance

~900M+
Total Confirmed TXs

~886K+
Blocks Mined (2009-2025)

~10 min
Average Block Time

~3,500
Avg TXs per Block

4 eras
Halving Epochs (so far)

Who Needs This Platform?

Two primary end-user profiles drive the OLAP requirements of this data mart.

Blockchain Analyst / Researcher

- Tracks TX volume trends across halving epochs
- Analyses fee rate distribution per block era
- Compares on-chain activity vs. market price
- Studies UTXO age bands and coin dormancy
- Monitors miner revenue (subsidy + fees)
- Identifies SegWit adoption growth over time

PRIMARY USER

Exchange Operator / Trader

- Monitors mempool congestion for fee estimation
- Correlates on-chain volume with price movements
- Audits large transaction flows (whale activity)
- Tracks confirmation times by fee tier segment
- Analyses historical block fullness vs. demand
- Evaluates market Fear & Greed vs. on-chain data

SECONDARY USER

Two Primary Data Sources

SOURCE 1

Bitcoin On-Chain Data

mempool.space API · blockchain.com/api · Bitcoin Core RPC

- Block height, hash, timestamp, size (bytes), weight
- Transaction count per block, total fees collected
- Individual TX: txid, input count, output count
- Fee amount (satoshis), fee rate (sat/vByte)
- Input/output values (satoshis), script type
- SegWit witness flag, virtual size (vBytes)
- Coinbase TX details - miner reward & address
- Mempool snapshot: pending count, avg fee rate

REST / JSON | CSV flat file | Free tier

SOURCE 2

Bitcoin Market Data

CoinGecko API · Yahoo Finance · Alternative.me (Fear & Greed)

- Date, open, high, low, close price (USD/OHLCV)
- 24h trading volume, market capitalisation (USD)
- Network difficulty & estimated hash rate
- Fear & Greed Index (daily score 0-100)
- Fear & Greed label: Extreme Fear → Extreme Greed
- Bitcoin dominance % vs. total crypto market cap
- NVT ratio - Network Value to Transactions
- Exchange-level volume breakdown (Binance, etc.)

REST / JSON | CSV export | Since 2009

Data Profile - What We Extract & Why

Dataset	Grain (unit of a row)	Est. Volume	Format	Update Rate	Our Scope
Block metadata	1 row per mined block	~886K rows	JSON → CSV	~10 min	Last 200K blocks
Transaction-level	1 row per TX in a block	~900M rows (subset)	JSON → CSV	~10 min	~50M TX sample
TX Inputs / Outputs	1 row per UTXO input/output	Multi-billion (filter)	JSON → CSV	~10 min	Sampled subset
Market OHLCV	1 row per day	~5,500 rows (2009–)	CSV / REST	Daily	Full history
Fear & Greed Index	1 row per day	~2,000 rows (2019–)	JSON / REST	Daily	Full history

📊 Operational Data

Raw TX & block data - close to generation, not pre-aggregated. Ideal for DW fact grain.

🔀 Heterogeneous Formats

Mix of REST JSON (live) + CSV flat files (historical). Validates SSIS multi-source ETL design.

🚩 Practical Scope

200K blocks ≈ 4 years of data (~700M TXs). Statistically representative while staying manageable.

Business Process, Dimensions & Star Schema Sketch

Business Process: Bitcoin Transaction Settlement - each on-chain confirmed transaction in a mined block. **Grain:** One confirmed transaction.

DIM_DATE

- date_key (PK · YYYYMMDD)
- full_date
- year, quarter, month_num, month_name
- day_of_week, week_num
- is_weekend
- is_holiday
- halving_era (1–4)
- days_since_halving
- unix_timestamp_day

DIM_BLOCK

- block_key (PK)
- block_height
- miner_coinbase_address
- block_size_bytes
- nonce
- block_hash
- block_weight_units
- difficulty
- tx_count_in_block
- is_epoch_boundary

DIM_TX_TYPE

- type_key (PK)
- is_coinbase (bool)
- segwit_flag (bool)
- input_count_bucket
- output_count_bucket
- primary_script_type
- is_rbf_signalling
- size_tier (S/M/L/XL)

DIM_MARKET

- market_key (PK)
- fear_greed_label
- fear_greed_bucket
- price_trend
- volatility_regime
- market_cap_tier
- btc_dominance_tier
- nvt_signal
- cycle_phase
- post_halving_months

FACT_TRANSACTION

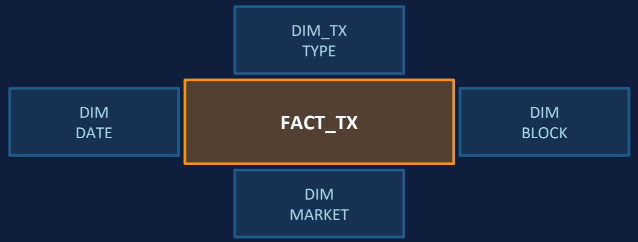
Grain: one confirmed Bitcoin transaction

Foreign Keys

- date_key → DIM_DATE
- block_key → DIM_BLOCK
- tx_type_key → DIM_TX_TYPE
- market_key → DIM_MARKET

Degenerate Dimensions

Star Schema - FACT_TRANSACTION at center



Measures (12 total)

fee_satoshis [SUM, AVG]
Total fee paid for the TX in satoshis

fee_rate_sat_vbyte [AVG, PERCENTILE]
Fee per virtual byte - miner priority signal

input_value_sat [SUM]
Total value of all UTXOs consumed as inputs

output_value_sat [SUM]
Total value of all outputs created

tx_size_bytes [AVG, MAX]
Raw byte size of the transaction on disk

tx_weight_units [AVG, SUM]
SegWit weight - used for fee rate calculation

btc_price_usd_close [SNAPSHOT]
Daily BTC/USD closing price on transaction day

btc_price_usd_high [SNAPSHOT]
Daily high BTC price - intraday range ceiling

btc_price_usd_low [SNAPSHOT]
Daily low BTC price - intraday range floor

volume_24h_usd [SUM]
Total 24h USD trading volume on transaction day

market_cap_usd [SNAPSHOT]
BTC market capitalisation on transaction day

fear_greed_score [AVG]
Raw 0-100 Fear & Greed score; categorical label lives in DIM_MARKET

Decision-Support Questions & Expected Outputs

N1	ANALYST	**How does avg fee rate evolve month-over-month across halving epochs?** Dims: DIM_DATE × DIM_BLOCK × DIM_TX_TYPE	*Line chart: fee trends with epoch markers, spike at congestion events*
N2	ANALYST	**What % of transactions are SegWit-native vs. legacy per quarter?** Dims: DIM_DATE × DIM_TX_TYPE	*Stacked bar: SegWit adoption growth 2017 → present*
N3	TRADER	**Do on-chain TX volumes spike before large BTC price movements?** Dims: DIM_DATE × DIM_MARKET × DIM_BLOCK	*Dual-axis chart: daily TX volume vs. BTC price overlay*
N4	TRADER	**What is the fee rate distribution during high vs. low Fear & Greed index periods?** Dims: DIM_MARKET × DIM_TX_TYPE × DIM_DATE	*Box-plot / histogram: fee rates bucketed by market sentiment regime*
N5	ANALYST	**How does total miner revenue (subsidy + fees) change across halving epochs?** Dims: DIM_BLOCK × DIM_DATE	*Area chart: miner revenue trend with halving drop markers*

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PART 1 - ANALYSIS COMPLETE

What We've Established

- ✓ Domain: Bitcoin blockchain transaction analytics (on-chain + market)
- ✓ 2 data sources: mempool.space/blockchain.com API + CoinGecko/Alt.me
- ✓ 2 user profiles: Blockchain Analyst & Exchange Trader/Operator
- ✓ 1 business process: confirmed Bitcoin transaction settlement
- ✓ 4 dimensions: DIM_DATE, DIM_BLOCK, DIM_TX_TYPE, DIM_MARKET (categorical regime)
- ✓ 8-10 attributes per dimension - exceeds minimum requirement
- ✓ 12 measures: fee, fee rate, input/output value, TX size, weight + BTC price (OHLC), volume, market cap, Fear & Greed score, NVT ratio
- ✓ 5 OLAP user needs mapped to expected visualisations

NEXT → Part 2: Multidimensional model design · Logical Data Map (LDM) · ETL planning with SSIS